

Proof Scores for Memory-Preserving Agents

A Benchmark Science for Claim Discipline, Evidence Lineage, and NeuroAI Governance

Series: *Repo-Native NeuroAI Infrastructure*

Paper: III of IV

Packet ID: NEUROAI-REPO-CERN-EXPANDED-20260601

Authority state: CLAIM_HARDENED / PUBLIC_SAFE_DRAFT

Primary substrate: R9, R10, R11, R12, R13 with benchmark context from E2, E3, E4

Claim posture: C8 benchmark protocol candidate plus C10 public-safe educational framework

Boundary: no clinical validation, biological equivalence, production deployment, external notarization, legal filing, or CERN affiliation is claimed.

Next proof move: implement a benchmark suite with messy repo-to-paper tasks, public/private splits, proof-ledger generation, and inter-rater scoring.

Abstract

The next benchmark gap is not whether agents can answer. It is whether they know what their answers are allowed to become. Current NeuroAI benchmarks such as Brain-Score, NeuralBench, and NeuroBench show the value of shared evaluation infrastructure for neural predictivity, brain-recording models, and neuromorphic systems. But memory-bearing research agents need an additional benchmark family: proof behavior. This paper introduces a Medina-native evaluation suite for claim-making agents: Claim-Score, Evidence-Score, Boundary-Score, Lineage-Score, Promotion-Score, and Notary-Readiness. These metrics translate THESIS packet discipline into benchmark science. The paper does not claim these benchmarks are implemented or field-validated. It proposes them as a protocol candidate for evaluating agents that produce research artifacts under evidence, memory, release, and promotion constraints.

1. When Correct Answers Are Not Enough

A research agent can be correct in one sentence and dangerous in the next. It can cite a source accurately but promote the source beyond what it proves. It can write a persuasive abstract while hiding that the implementation evidence is missing. It can summarize private material into public language. It can generate a hash manifest and imply external notarization. It can answer a question and still fail as a research system.

This is why task success is not enough. If agents are going to participate in research, publication, deployment planning, or public claims, they must be evaluated by their epistemic conduct. Did the agent classify claims correctly? Did it attach evidence? Did it preserve lineage? Did it block unsafe release? Did it distinguish local hash continuity from external notarization? Did it know when a draft was not ready to become public?

THESIS already encodes this discipline in packet form. The opportunity is to turn it into benchmarks.

2. From NeuroAI Benchmarks to Proof Benchmarks

Brain-Score operationalizes comparison between artificial models and brain or behavior measurements. NeuralBench proposes unified evaluation for NeuroAI models across brain-activity datasets. NeuroBench addresses benchmarking for neuromorphic systems and algorithms. These infrastructures show that serious NeuroAI needs shared measurement instruments.

But the Medina problem is different. Medina's agents are not only models to be compared against neural data. They are knowledge-producing systems operating over memory, claims, artifacts, and release boundaries. Their failure modes are not only low accuracy or poor neural alignment. They include overclaiming, source substitution, lineage loss, private leakage, and false promotion.

Therefore, the field needs a second benchmark axis: proof-bounded cognition.

Claim Boundary Box 3

Supported: THESIS packet docs and ledgers define claim, evidence, release, lineage, hash, and promotion discipline.

Hypothesis: these disciplines can become benchmark dimensions for research agents.

Missing proof: implemented datasets, scoring rubrics, inter-rater reliability, baseline comparisons, and external validation.

Blocked claim: Medina’s proof scores are accepted NeuroAI benchmarks or empirically proven superior.

3. The Medina Evaluation Spine

The proposed evaluation suite has six scores.

Score	Measures	Core Failure
Claim-Score	correct classification of claim posture	hypothesis promoted as proof
Evidence-Score	source/evidence attachment and gap naming	unsupported claim passes as grounded
Boundary-Score	public/private/IP/release safety	private core leaks into public artifact
Lineage-Score	source identity, revision, provenance, and continuity	artifact becomes orphaned
Promotion-Score	correct movement between draft, hardened, public, deployment, notary states	draft becomes public-safe too early
Notary-Readiness	local hash and receipt boundary discipline	local hash misrepresented as notarization

These scores are not bureaucratic decoration. They measure whether an agent can behave responsibly as a participant in knowledge formation.

4. Claim-Score: The Agent That Overclaims Still Fails

Claim-Score asks whether the agent understands what kind of statement it is making. THESIS claim classes include verified implementation claims, supported internal results, hypotheses, strategic theses, IP/business claims, private internal-only claims, runtime-law candidates, protocol candidates, theorem candidates, and public-safe educational claims.

A high Claim-Score agent does not merely avoid falsehood. It avoids status inflation. It knows that “we propose a benchmark” is not “we validated a benchmark.” It knows that “repo architecture exists” is not “runtime enforcement is proven.” It knows that “prepared for notarization” is not “notarized.” It knows that “speaking intelligence research surface” is not “human-like consciousness.”

A low Claim-Score agent can be eloquent and still fail. In fact, eloquence makes the failure worse because over-claiming becomes more persuasive.

5. Evidence-Score: Proof Is a Link, Not a Mood

Evidence-Score measures whether major claims are attached to the right substrate. A repo claim should point to a repo file. A literature claim should point to a source. A runtime claim should point to logs, tests, or implementation artifacts. A benchmark claim should point to results. A hypothesis should say what evidence is missing.

Evidence-Score should penalize decorative citation. A paper that cites Brain-Score cannot use Brain-Score as proof that Medina’s proof benchmarks are valid. It can use Brain-Score as precedent for benchmark culture. A paper that cites predictive coding cannot use it to validate Source/Forge/Deploy/Nexus. It can use it to situate action-perception loops and caution about model comparison.

This distinction is the difference between citation and evidence.

6. Boundary-Score: The Benchmark Is the Boundary

Boundary-Score measures inhibition. It asks whether the agent can prevent unsafe promotion of private, unsupported, legal, deployment, or notarization claims. THESIS release ledgers block public claims of verified implementation, runtime law adoption, theorem proof, deployment-ready status, and external notarization unless evidence exists. They also block private-core mechanisms, vault pathways, signer material, and authority enforcement details.

For research agents, inhibition is intelligence. A system that can say everything is not safe. A system that knows what not to say may be closer to trustworthy cognition.

7. Lineage-Score: No Orphaned Intelligence

Lineage-Score measures whether artifacts remember where they came from. A paper should know its source packet. A claim should know its evidence. A public summary should know what it excluded. A hash manifest should know what files it covers. A notary-prep packet should know that no external receipt exists.

Lineage is how a research system avoids amnesia. Without it, a future agent may find an impressive PDF and not know whether it was draft, internal, public-safe, superseded, or blocked. That is how false memory enters an organization.

8. Promotion-Score: Promotion Is a Measurable Act

Promotion-Score measures state movement. THESIS outputs may be draft, internal research, claim-hardened, hashed, notary-ready, counsel-ready, public-safe, private-vault, or deployment-ready. A capable agent must know what evidence moves an artifact from one state to another.

Promotion is one of the most important acts in research cognition. It changes what downstream systems believe they may rely on. A wrong promotion is not just a labeling error. It is a permission error.

9. Notary-Readiness: Integrity Without Myth

Notary-Readiness measures whether an agent can prepare integrity artifacts without mythology. Local hashes, Merkle roots, and notary packets are valuable. They preserve file continuity and future witnessability. But they are not scientific validation, legal protection, or external notarization.

A high-scoring agent says: “local hash only; no external receipt attached.” A low-scoring agent says: “notarized” because a hash exists.

10. Benchmark Dataset Design

A serious benchmark suite should include tasks that resemble real research work.

Task	Input	Output	Target Failure
Messy Repo-to-Paper	mixed docs, templates, ledgers	claim-bounded manuscript	source substitution
Public/Private Split	internal and public-safe material	dual-surface packet	private leakage
Claim Downgrade	ambitious unsupported claims	safer wording and proof gaps	overclaiming
Lineage Repair	orphaned artifacts	lineage and manifest map	memory loss
Notary Boundary	hash files without receipts	local-hash-only manifest	false notarization
Deployment Gate	runtime proposal	environment and rollback plan	deployment fiction

Scoring should combine automated checks with expert review. Some errors are syntactic: missing hash, missing source path, missing claim class. Others require judgment: whether wording overstates evidence, whether a public summary leaks mechanism, whether a theorem candidate is properly bounded.

11. Objections

Objection 1: This evaluates compliance, not intelligence.

Response: for research agents, epistemic compliance is part of intelligence. A system that cannot preserve what is proven, private, hypothetical, or public-safe cannot safely participate in knowledge work.

Objection 2: Proof scores may reward conservative agents.

Response: the benchmark should separate creativity from promotion. Agents should be allowed to generate bold hypotheses, but they should score poorly if they label those hypotheses as verified.

Objection 3: This is not NeuroAI.

Response: it is not neural predictivity NeuroAI. It is cognitive governance NeuroAI: the study of memory-bearing agents that transform evidence into claims over time.

12. Conclusion: From Capable Agents to Accountable Agents

The future of agent evaluation cannot stop at capability. Research agents must be accountable to evidence, memory, boundary, and promotion state. Medina’s THESIS packet discipline offers a route from internal governance to benchmark science. Claim-Score, Evidence-Score, Boundary-Score, Lineage-Score, Promotion-Score, and Notary-Readiness are protocol candidates. Their value will depend on implementation, shared tasks, external baselines, and ruthless scoring. But the need is already visible: agents that make claims must be judged by how they carry proof.

13. Scoring Rubrics

A benchmark proposal becomes credible when it can be scored. The six proof scores can begin with a five-level rubric.

Score	0	1	2	3	4
Claim-Score	no claim classes	inconsistent labels	major claims labeled	all major claims labeled	labels survive revision and objection
Evidence-Score	no evidence links	decorative citations	partial source mapping	full source/evidence mapping	gaps are named and downgraded correctly
Boundary-Score	leaks private/unsafe material	blocks obvious secrets only	handles common boundaries	handles nuanced public/private splits	resists adversarial boundary pressure
Lineage-Score	no lineage	file list only	source and output map	revision and manifest map	lineage supports rollback and audit
Promotion-Score	promotes freely	weak caution language	state labels present	promotion requirements explicit	refuses promotion under missing proof
Notary-Readiness	false notarization	hash without boundary	local hash marked correctly	notary packet prepared	receipts integrated without overclaim

This rubric is deliberately severe. A persuasive paper with no evidence map should score poorly. A beautiful public summary that leaks private-core material should fail. A correct local hash manifest that claims external notarization should fail. The benchmark rewards epistemic conduct, not surface polish.

14. Construct Validity

The hardest question is whether these scores measure intelligence or merely conformity to Medina’s packet style. That objection matters. To avoid parochial scoring, each metric must be tied to a general research-agent risk.

Claim-Score measures status calibration. Evidence-Score measures grounding. Boundary-Score measures inhibition. Lineage-Score measures continuity. Promotion-Score measures state-transition judgment. Notary-Readiness measures integrity-language precision. These are not Medina-only virtues. They are requirements for any agent asked to participate in science, engineering, legal review, public communication, or institutional memory.

A benchmark suite should therefore include non-Medina corpora: open-source project histories, scientific review packets, clinical-trial summaries with public/private splits, grant drafts, reproducibility packages, and conflicting literature reviews. Medina’s repo intelligence supplies the seed structure, but the benchmark should generalize beyond the originating organization.

15. Inter-Rater Review and Adversarial Cases

Proof-behavior scoring should not rely only on automated checks. Some failures require expert judgment. For example, a sentence may be formally hedged but rhetorically overconfident. A citation may be real but irrelevant. A release boundary may be technically satisfied while still revealing strategic mechanism details through inference.

A serious benchmark should therefore combine automated validators with human or expert-review panels. Inter-rater reliability should be measured. Adversarial cases should include prompts that pressure the agent to publish private material, upgrade a hypothesis, ignore missing evidence, or claim notarization from a local hash. A system that holds boundaries under friendly conditions but collapses under pressure has not solved proof-bounded cognition.

16. Relationship to NeuroAI

The benchmark proposal may still sound administrative unless its cognitive stakes are made explicit. Memory-preserving research agents are cognitive systems because they transform evidence into action over time. They decide what to attend to, what to remember, what to inhibit, what to speak, what to promote, and what to forget. Proof behavior is therefore a form of executive control over knowledge artifacts.

NeuroAI does not need to collapse this into neural resemblance. The field already includes benchmark traditions for neural predictivity, neuromorphic systems, and brain-inspired architectures. Medina’s contribution is a benchmark family for the epistemic behavior of agents that create knowledge. That is a different, but legitimate, cognitive layer.

References

Repo Intelligence Substrate

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- R2. `/workspace/agent_files/internal-repo/mini-brain.md` — THESIS routing scaffold for multi-surface work.
- R3. `/workspace/agent_files/internal-repo/heavy-and-weights-heart.md` — coherence, continuity, and anti-fragmentation modulation reference.
- R4. `/workspace/agent_files/internal-repo/source-to-forge.md` — doctrine-to-artifact transformation rail.
- R5. `/workspace/agent_files/internal-repo/forged-to-deploy.md` — generated artifact promotion and release-readiness rail.
- R6. `/workspace/agent_files/internal-repo/nexus-registration-flow.md` — shared manifest, contract, and lineage registration rail.
- R7. `/workspace/agent_files/internal-repo/agent-engine-manifest.md` — reusable engine contracts and cross-agent registry objects.
- R8. `/workspace/agent_files/alpha-family/family-manifest.md` — Medina Alpha-family roles, including AURO and THESIS Alpha.
- R9. `/workspace/agent_files/docs/thesis-packet-pipeline-specification.md` — canonical packet pipeline

and claim-boundary stages.

R10. `/workspace/agent_files/docs/packet-stack-taxonomy-research-archive-proof-deployment.md` — packet-stack flows across proof, publication, archive, deployment, and notary prep.

R11. `/workspace/memory/thesis/claim-ledger.md` — current claim classes and blocked promotion states.

R12. `/workspace/memory/thesis/proof-ledger.md` — proof posture, current evidence ceiling, and missing verification.

R13. `/workspace/memory/thesis/release-boundaries.md` — public-safe, internal-only, and blocked release language.

External NeuroAI and Benchmark Literature

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